

## Instructor comments

This exam information page is intended to provide approximate information about the content weighting and the types of questions you can expect on the final exam. **I will not be answering further questions about which formulas you should or should not memorize, or which derivations you should or should not know.**

I genuinely want you to engage with the course material rather than skipping over topics simply because they may not appear directly on an exam. In order to review the material calmly and not worry when a larger derivation appears that you may not understand right away, please give yourself enough time to study and review the material.

But if you do end up preparing for this exam with limited time, you can use this exam sheet in order to focus your efforts better.

## Breakdown of points on Exam

50 marks in total. Front page of the exam has been published.

- Multiple choice (12 marks)
- Question 1 (5 marks)
- Question 2 (8 marks)
- Question 3 (8 marks)
- Question 4 (7 marks)
- Question 5 (10 marks)

## A note on formulas to memorize

- You are given all the formulas on the aid sheet.
- You are not asked to memorize giant bias expressions etc..
- You are expected to know "formulas" that aren't on the sheet and are considered intuitive. Example, proportional allocation, Neyman allocation, inclusion probabilities, a class-weighting adjustment estimate of  $\hat{\phi}_c$ , etc...

## Approximate Topic Weights

The following percentages are intended only as a rough guide to the relative emphasis of topics on the final exam.

| Topic                                     | Approximate Weight |
|-------------------------------------------|--------------------|
| Simple Random Sampling (SRS)              | 10%                |
| Stratified Sampling                       | 15-20%             |
| Cluster and Multi-stage Sampling          | 15-20%             |
| Unequal Probability Sampling              | 10-15%             |
| Nonresponse and Missing Data              | 20-25%             |
| Survey Design and Selection Bias          | 10-15%             |
| Ratio & Regression Estimation             | 5%                 |
| Domain Estimation and Post-stratification | 5-10%              |
| Observational Studies & Causation         | 2-5%               |

## Types of Questions You Should Expect

The following percentages are intended only as a rough guide to the relative emphasis of skills tested on the final exam. Throughout the exam, you may be asked to:

### Define (5–10%)

Examples:

- Sampling frame
- Target population
- MCAR, MAR, and NMAR

### Explain (20–25%)

Examples:

- Why and when to use certain sampling methods
- Why different estimation methods may be beneficial (e.g. ratio estimators)

### Justify (15–20%)

Examples:

- Choice of sampling design
- Choice of observation allocation strategy
- Choice of auxiliary variables

### Derive (20–25%)

Examples:

- Properties of estimators
- Sample size derivations

### **Calculate (20–25%)**

Examples:

- Point estimates
- Selection probabilities

### **Interpret (10–15%)**

Examples:

- Confidence intervals
- Missing data mechanisms

### **Critique (5–10%)**

Examples:

- Survey questions
- Sampling designs
- Sources of bias